

Product Requirements Document

Professional-Skepticism Engine — a Multi-Agent Audit Investigation Crew

Field	Value	Field	Value
Product	Professional-Skepticism Engine	Doc type	Product Requirements Document
Version	1.1	Status	Reviewed — open items resolved
Owner	Capstone Team 01 (2 members)	Date	2026-06-25
Target release	5 working days	Pattern	Multi-agent (debate + verification)

1. Overview

The Professional-Skepticism Engine is a **multi-agent audit investigation system**. For each flagged transaction or piece of audit evidence, a crew of role-specialised Claude agents **debates and verifies** the case the way a real audit team does: an Evidence agent assembles the facts, a **Challenger** argues it is a problem, a **Defender** argues it is legitimate, an **Adjudicator** weighs both and assigns a risk + confidence + disposition, and a **Verifier** checks that the conclusion is grounded in evidence. Uncertain or material cases pause for a human reviewer; every decision and the full debate transcript are audited. The design directly operationalises **professional skepticism** (ISA 200) and the preparer → reviewer → partner review hierarchy.

2. Background & Problem Statement

A single LLM asked to judge a flagged item in one prompt is prone to **confirmation bias and hallucination** — it tends to find one plausible story and commit to it. In audit that is unacceptable: conclusions must be challenged, defended, and independently verified before they are relied upon. Rule-based detection is cheap and deterministic but cannot exercise judgement; a lone LLM exercises judgement but without challenge or verification. The product closes this gap with an **adversarial, role-separated multi-agent design** in which the independence of the agents is what produces a defensible verdict.

3. Why Multi-Agent Is Mandatory (not optional)

This is the core design rationale — the project is built around a pattern that **cannot be collapsed into a single prompt or function** without losing its value:

- **Independence is the mechanism.** The Challenger and Defender must reason *separately*; if one model played both roles it would rationalise a single narrative. Separation is what surfaces the counter-arguments.
- **Verification needs a different agent.** A model checking its own output for hallucination is unreliable; an independent Verifier with a distinct objective catches ungrounded claims and forces a revision.
- **Dynamic control.** The Supervisor re-plans and re-runs the debate when verification fails — a control loop, not a linear script.
- **Empirically testable.** §18 specifies an A/B evaluation (single-prompt vs. the crew) that *measures* the multi-agent uplift — proving the agents earn their place rather than asserting it.

Defensible claim: “We use multiple agents because the independent, adversarial reasoning measurably reduces false positives and catches hallucinations a single pass misses. Remove any agent and the verdict degrades — which we demonstrate empirically.”

4. Goals & Non-Goals

4.1 Goals

- **G1** — Produce a challenged, defended, and verified risk verdict for each case, with citations.
- **G2** — Demonstrate a measurable quality uplift of the multi-agent crew over a single-prompt baseline.
- **G3** — Keep a human durably in the loop for uncertain / material cases.
- **G4** — Make every verdict grounded (RAG policy + external data) and fully auditable, including the debate transcript.

4.2 Non-Goals (this release)

- Production ERP/SAP connectors; OCR; multi-tenant; RBAC/SSO at scale.
- Autonomous remediation / posting to the ledger; the engine recommends, humans decide.
- Large-scale fine-tuning; we use prompting + orchestration only.

5. Success Metrics (KPIs)

Metric	Definition	Target (MVP)
Multi-agent uplift	Crew vs. single-prompt: change in false-positive rate on a labelled set	≥ 20% FP reduction
Grounding rate	Adjudications passing the Verifier on first or second pass	≥ 95%
Hallucination catch	Ungrounded claims caught by the Verifier ÷ injected	≥ 80% on a seeded test
Auto-clear rate	High-confidence, verified, low-risk cases auto-cleared	30–50%
Reviewer effort	Minutes per case vs. manual baseline	≥ 50% reduction
Durability	Runs resumable after restart / mid-debate	100%

6. Personas & Stakeholders

Persona	Role / need
Audit Senior / Reviewer	Primary user; reviews the debate + disposition, approves / rejects with accountability.
Engagement Manager / Partner	Escalation approver for high-risk; owns sign-off.
Capstone Dev Team (2)	Build & operate; need modular, testable agents.
Methodology / Risk (future)	Owns the policy KB and the confidence / materiality thresholds.

7. User Stories

ID	User story
US-1	As a reviewer, I want each case argued from both sides so I can trust the verdict isn't one-sided.
US-2	As a reviewer, I want the AI's conclusion verified against evidence so I'm not misled by a confident hallucination.
US-3	As a reviewer, I want to read the full debate transcript and approve / reject with a comment and signature.
US-4	As a partner, I want high-risk / material cases escalated to me automatically.
US-5	As compliance, I want an immutable trail of the debate, decision, and sources for every case.
US-6	As a developer, I want the run to survive a restart mid-investigation.

8. Scope

In scope (MVP, 5 days)	Out of scope (future)
GL-flagged items / synthetic audit cases as input	OCR, ERP/SAP connectors, live client data
6 role-specialised Claude agents + debate + verify loop	Many specialist agents; production fan-out at scale
Policy RAG over a small company-policy KB	Enterprise knowledge graph; auto policy ingestion
Confidence-gated human review (interrupt/resume)	RBAC/SSO, multi-tenant, e-signature platform
A/B eval vs single-prompt; SQLite + Streamlit	Kubernetes, autoscaling, online learning

9. System Architecture & End-to-End Workflow

A LangGraph state machine orchestrates the crew. The Supervisor routes; the Evidence agent grounds; the Challenger and Defender debate; the Adjudicator decides; the Verifier gates on grounding and can trigger a re-run. Diagram in Appendix A.

#	Step	What happens / artifact
1	Case intake	A flagged GL item / evidence becomes one investigation case.
2	Supervisor · Planner	Decompose the case, decide which specialists to run, set escalation.
3	Evidence agent	RAG policy + history + live data → evidence{} with citations.
4a	Challenger	Argues fraud / error / control failure; cites red flags → challenger_view.
4b	Defender	Argues the legitimate business rationale → defender_view.
5	Adjudicator	Weights both → adjudication{risk · confidence · disposition · citations}.
6	Verifier · QA	Checks every claim is grounded; if not, returns to the Adjudicator / Supervisor to revise.
7	Confidence gate	≥0.90 & verified & Low → auto-clear · 0.70–0.90 → review · else manual.
8	Human review	interrupt(): reviewer sees the debate transcript, approves / rejects / signs → resume().
9	Report & Audit	Risk-ranked case file (debate trail + citations) + immutable audit log.
*	Persistence	SQLite checkpointer persists state & transcript at every step — durable + resumable.

10. Agent Specifications

Agent	Role & why it must be separate	Key artifacts
Supervisor · Planner	Decomposes the case, routes to specialists, decides escalation & re-runs. A control agent, not a step.	plan · routing · escalation decision
Evidence agent	Tool-using retrieval (RAG + history + APIs); separated so judgement never does its own data-gathering.	evidence{} + citations · api_metadata
Challenger	Independent skeptic; argues the worst case. Must be isolated from the defence to avoid one-sided reasoning.	challenger_view (claims · red flags)
Defender	Independent counter-argument; surfaces benign explanations — this is what cuts false positives.	defender_view (rationale)
Adjudicator	Neutral judge; weighs both sides, assigns risk + confidence + disposition, must cite. Cannot be an advocate.	adjudication{risk · confidence · disposition · citations}
Verifier · QA	Independent grounding check; catches hallucination and forces revision. A model can't reliably check itself.	verification{passed · issues}
Human reviewer	Accountability for material / uncertain cases; durable interrupt.	review_history[] (action · comment · signature)

11. Functional Requirements

MoSCoW priorities (**Must** / **Should** / **Could**).

11.1 Orchestration & Evidence

ID	Requirement	Priority
FR-OR-1	The Supervisor must orchestrate the crew over a typed, checkpointed state and route per case.	Must

ID	Requirement	Priority
FR-OR-2	On Verifier failure, the Supervisor re-runs the debate / adjudication up to a maximum of 2 rounds; if it still fails, the case auto-escalates to a human reviewer.	Must
FR-EV-1	The Evidence agent must assemble an evidence pack from Policy RAG + transaction history + ≥ 1 external source, with citations.	Must
FR-EV-2	External calls must time out and fall back to cache; missing evidence is recorded, not fatal.	Must

11.2 Debate, Adjudication & Verification

ID	Requirement	Priority
FR-CH-1	The Challenger must produce a structured case-for-concern (claims + cited red flags) from the evidence only.	Must
FR-DF-1	The Defender must produce a structured case-for-legitimacy (rationale) independently of the Challenger.	Must
FR-AD-1	The Adjudicator must weigh both views and output risk_level + confidence + disposition + citations.	Must
FR-AD-2	The Adjudicator must cite the specific evidence / policy clause behind its decision.	Must
FR-VE-1	The Verifier must check each adjudicator claim is supported by retrieved evidence and flag unsupported ones.	Must
FR-VE-2	If verification fails, the case must loop back for revision (bounded), not pass silently.	Must

11.3 Decision, Human review, Report & Audit

ID	Requirement	Priority
FR-CG-1	Route by confidence + verification + materiality: auto-clear / reviewer / manual.	Must
FR-HR-1	Review / manual cases pause via interrupt() with state + transcript persisted.	Must
FR-HR-2	Reviewer sees the full debate transcript + disposition and approves / rejects / comments / signs.	Must
FR-HR-3	Material / high-risk cases auto-escalate to a partner; SoD enforced (preparer $\neq$ approver).	Should
FR-RP-1	Produce a risk-ranked case file (debate trail + citations) in Markdown / HTML.	Must
FR-AU-1	Append-only audit log of the debate, decision, reviewer, and sources.	Must
FR-UI-1	Streamlit UI: case list, debate-transcript view, review queue with approve / reject / resume.	Must

12. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Reliability	Checkpointed state + transcript; resume mid-debate after restart; bounded re-runs; cache fallback.
NFR-2	Performance	Process a case (full debate) in $< \sim 30$ s typical; batch of 50 demo cases in < 20 min.
NFR-3	Cost	Cheap rules pre-filter; cap debate rounds; cache evidence; route model size per role.
NFR-4	Security	API keys in secret store; PII not logged in clear; least-privilege.
NFR-5	Observability	Per-agent traces (tokens, latency, cost) + run metrics; LangSmith optional.
NFR-6	Auditability	Immutable trail incl. full debate transcript; citations on every verdict.
NFR-7	Maintainability	Role prompts as versioned templates; unit + eval tests; README; Docker.

13. RAG & Policy Grounding

The Evidence agent retrieves company-specific context so the debate argues against *real* rules, not model priors.

- **KB contents:** approval / authority matrices, SOPs, procurement policy, related-party policy, audit standards (ISA), vendor-master rules, historical audit workpapers, and master data.
- **Ownership:** the KB is maintained jointly by Audit Methodology, Finance, Procurement, and Master Data teams; policy updates are a re-index (data change), not a code change.
- **Pipeline:** chunk policies by clause → embed → vector DB (Chroma / FAISS) with metadata (doc · clause · version) → hybrid search + re-rank at query time.
- **Citation:** every retrieved clause carries an id; the Adjudicator must cite it — e.g., “breaches approval limit AP-02 §1”.

14. Decision & Confidence Logic

Condition	Route
confidence $\geq$ 0.90 AND Verifier passed AND risk = Low AND amount < materiality	Auto-clear (QA-sampled)
$0.70 \leq$ confidence < 0.90 (and verified)	Reviewer queue
confidence < 0.70 OR Verifier failed twice OR risk = High OR amount $\geq$ materiality	Manual / partner escalation

Materiality threshold = USD 25,000. Mandatory partner review is triggered by **any** of: amount $\geq$ USD 25,000, High risk, or failed verification. Confidence thresholds (0.90 / 0.70) and materiality are configuration owned by Audit Methodology / Risk.

15. Data Model — InvestigationState

```
case · evidence{} · debate_transcript[] · challenger_view · defender_view ·  
adjudication{risk·confidence·disposition·citations} · verification{passed·issues} ·  
routing · review_history[] · audit_history[] · report
```

16. Technology Stack & External APIs

16.1 Technology stack

Layer	Technology	Purpose
Orchestration	LangGraph (StateGraph + SQLite checkpoint, interrupt/resume)	Durable multi-agent control loop
LLM	Anthropic Claude via langchain-anthropic	Sonnet for the 6 reasoning agents (temp per role); Haiku for report generation
Schemas	Pydantic v2	Typed state + per-agent structured outputs
RAG / vector DB	Chroma or FAISS + embeddings	Policy-clause retrieval (hybrid search)
Real-time HTTP	httpx	Evidence agent external calls
Persistence	SQLite	Checkpoints, debate transcript, audit log
UI	Streamlit	Case list, debate viewer, review queue
Eval / tests	pytest + a labelled case set	A/B eval + unit / mocked-HTTP tests
Packaging / obs	Docker · LangSmith (optional)	Reproducible runtime; per-agent traces

16.2 Third-party APIs called

API / service	Category	Purpose	Scope
Anthropic Claude	LLM	Every agent (debate, adjudicate, verify)	MVP
Frankfurter / exchangerate.host	Currency / FX	Historical FX evidence	MVP
Google Shopping / Open Food Facts	Market price	Price-reference evidence	Future
GST / GSTN verify	Vendor / Tax	Vendor / tax validity evidence	Future
Local cache	Resilience	Offline fallback for the demo	MVP

17. Evaluation Plan (proving the multi-agent uplift)

- Build a small **labelled set** (e.g., 30–50 cases with known fraud / legitimate ground truth) from the synthetic GL data.
- Run each case through (A) a **single Claude prompt** baseline and (B) the **multi-agent crew**.
- Compare false-positive rate, false-negative rate, and grounding (citation) quality.
- Inject ungrounded claims to measure the **Verifier's hallucination-catch rate**.
- Report the delta — this is the headline evidence that the multi-agent design is necessary, not decorative.

18. Assumptions, Dependencies & Constraints

- **Assumptions:** synthetic / sample audit cases are acceptable; a small policy set seeds the KB; an Anthropic key is available.
- **Dependencies:** Claude API; LangGraph/Streamlit stack; FX API (cache-mitigated).
- **Constraints:** 2 developers, 5 working days; debate rounds capped for cost; confidence is model-reported (not calibrated).

19. Risks & Mitigations

Risk	Mitigation
Debate adds latency / cost	Cap rounds; pre-filter with rules; cache evidence; route model size per role
Agents collude / converge (lose independence)	Strict role prompts; the Defender never sees the Challenger's reasoning until adjudication
Verifier misses hallucination	Require citation ids; seed ungrounded-claim tests; bounded re-run
Scope creep in 5 days	MoSCoW + cut-list; the eval and HITL are the protected headlines
LLM cost overrun	Round caps, caching, small labelled set, spend alert

20. Milestones & Timeline (5 working days, 2 people)

Day	Milestone	Owner focus
1	Contracts (InvestigationState) + LangGraph skeleton + 1 Claude agent working + dataset	A: orchestration · B: data + UI shell
2	Evidence agent + Policy RAG; Challenger + Defender agents	A: agents · B: RAG / policy KB
3	Adjudicator + Verifier loop + confidence routing	A: reasoning loop · B: debate-transcript UI
4	Human review (interrupt/resume) + audit log + case report	A: HITL · B: report / dashboard

Day	Milestone	Owner focus
5	A/B eval (single-prompt vs crew) + integration + demo / slides	Both

21. Acceptance Criteria (Definition of Done)

- A case runs end-to-end: evidence → Challenger & Defender debate → Adjudicator → Verifier → verdict.
- Verifier demonstrably sends at least one ungrounded case back for revision.
- At least one case auto-clears and one routes to a human, who approves / rejects with a signed comment; graph resumes.
- The A/B eval shows a measurable improvement of the crew over the single-prompt baseline.
- A run survives a restart mid-debate; `pytest` passes; `docker run` works.
- 5-minute demo: live case + the debate transcript + the eval result.

22. Resolved Design Decisions

Items that were open during drafting are now decided:

Question	Decision
Debate rounds before escalation	Maximum 2 debate rounds. If the Verifier still rejects the adjudication, the case auto-escalates to a human reviewer.
Materiality for mandatory partner review	Transactions ≥ USD 25,000, high-risk cases, or cases with failed verification require mandatory partner review.
KB seed & ownership	Seeded with approval matrices, SOPs, procurement & related-party policies, audit standards (ISA), vendor-master rules, historical workpapers, and master data. Owned by Audit Methodology, Finance, Procurement, and Master Data.
Model per role	MVP: Claude Sonnet for all reasoning agents, Claude Haiku for report generation. Future: model routing — larger models for complex reasoning, smaller for reporting, to optimise cost / performance.

Appendix A — Architecture Diagram

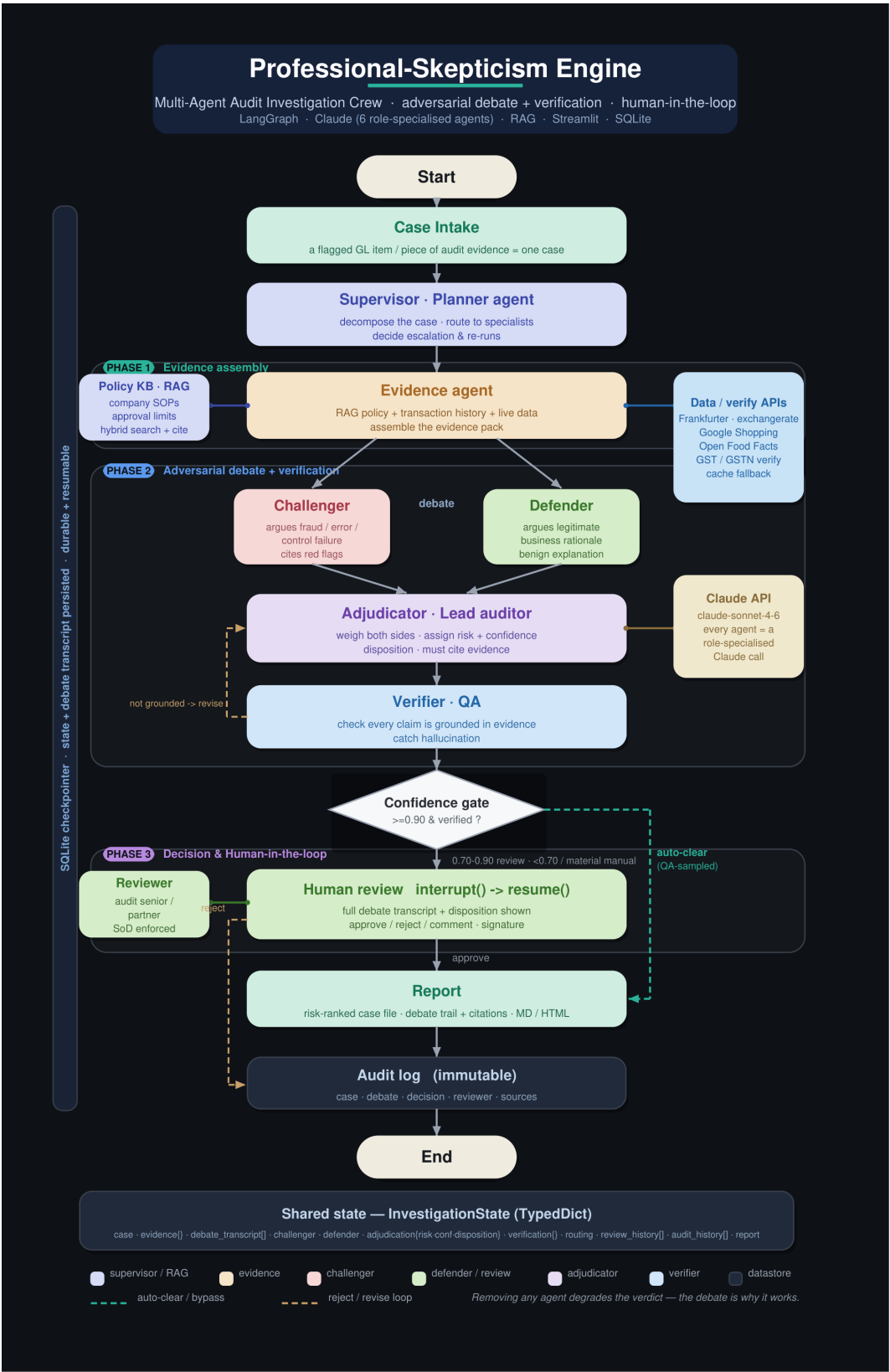


Figure 1 — The investigation crew. Evidence feeds an adversarial Challenger/Defender debate; the Adjudicator decides; the Verifier gates on grounding and loops back on failure; the confidence gate routes to auto-clear or the human-review interrupt. Every agent is a role-specialised Claude call; SQLite persists state + transcript.

Appendix B — Glossary

Term	Meaning
Professional skepticism	Audit mindset of questioning and critically assessing evidence (ISA 200)
Adversarial debate	Independent agents argue opposing positions to surface stronger reasoning
Adjudicator	The neutral judge agent that decides risk / confidence / disposition
Verifier	Independent QA agent that checks claims are grounded in evidence
interrupt() / resume()	LangGraph primitives that durably pause and continue the graph
InvestigationState	The shared typed state (incl. the debate transcript) passed between agents
Materiality	Amount threshold above which a case always requires human / partner attention